

Ryan Duong

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EDUCATION

Brown University

Providence, RI

Bachelor of Science in Applied Mathematics & Computer Science (4.00 GPA)

Expected May 2028

Coursework: (Graduate) Advanced Computer Graphics, Computational Neuroscience, Computer Vision, Information Theory, Computer Systems, Probability & Statistics, Numerical Optimization, Data Structures & Algorithms

Activities: Model UN (Secretary-General), CS Undergrad Group (E-Board), Brown Space Engineering

TAing: Accelerated Intro to CS (Fall 2025), Computer Architecture (Spring 2026)

EXPERIENCE

Software Engineer Intern, Machine Intelligence

June 2026 - August 2026

Roblox

San Mateo, CA

- Living a childhood dream.

Software Engineer Intern

June 2025 - August 2025

HyAxiom

East Hartford, CT

- Shipped HyAxiom's **first** mobile app tracking live analytics for **1500+** powerplants, using **React Native & Expo**.
- Redesigned cloud pipelines handling **700+** datapoints w/**AWS Lambda & CloudFormation** to reduce overhead.
- Expanded internal APIs for both **MySQL** & **NoSQL S3** databases, cutting response latency from **340ms to 140ms** w/fetching optimizations and **Redis** caching of previous real-time datapoints.
- Coordinated cross-functionally w/electrical & sales teams via **Agile/Scrum** sprints to develop high-priority features.

Undergraduate Researcher

January 2026 - Present

Brown University Interactive 3D Vision & Learning Lab

Providence, RI

- Generalizing robotic manipulation of deformables using **world action models** based on **point-flow motion**.
- Created a procedural scene generator to expand pretraining on **PTv3 Transformer** models, increasing accuracy by **15%**

Indie Game Developer

July 2017 – Present

Lugical

South Windsor, CT

- Scaling an indie game studio on Roblox to **60M+** plays, **2.5K+** concurrent users (CCU), **\$10K+ ARR**, & **5% CTR**
- Scripting game-logic, user interactions, client-server networking, & rendering optimizations, reducing ping by around 65%
- Engaged w/Roblox product teams to represent developer interests, as **1 of 300** Community Feedback Program members.
- **Recipient of the Global Citizenship Award @ Roblox Inspire 2024 Challenge.**

PROJECTS

Multi-Agent Simulations for AI Safety 🦾 | *Node.js, Socket.io, React, Matplotlib, Cerebras, Llama 5*

- Architected a **social game environment** acting as a **high-pressure environment** for LLM agents.
- Drafted a 4-person report **✍** finding psychological traits (deceit, fear, etc.) reduced decision-making outcomes by **64.5%**.
- Analyzed how individual **AI misalignment** aggregates into emergent social structure w/**multi-channel behavioral data** (reasoning traces, CoT, public/private communication)

3D Scene Reconstruction w/Smart Glasses 🦾 | *MentraOS, Supabase, Railway, PyTorch, NumPy, ngrok*

- **Won 2nd Place @ HackMIT 2025 (Mentra Challenge)**
- Crafted a **3D environmental reconstruction** pipeline for **smart glasses** using camera capture & vision transformers.
- Built a voice-activated app to control camera recording, caching 2D image scans into **Supabase** for AI inference.
- Fine-tuned VGGT & Segment Anything to construct point-cloud scenes for potential use in forensic analysis & gaming.

Cross-Platform, Customizable Video Game Camera 🦾 | *Lua, Rojo*

- Designed real-time raycasting for customizable, smooth, **3D/2D camera motion** not natively supported on Roblox.
- **Open-sourced** to the community as a direct alternative to Roblox's w/**1000+** installs on GitHub & Roblox Marketplace.

TECHNICAL SKILLS

Languages: C++, C, Python, TypeScript, JavaScript, Lua, SQL

Developer Tools: Git, Docker, Redis, ngrok, Jupyter, AWS, MongoDB, Unity, Unreal Engine

Frameworks: React, React Native, Expo, Flask, Next.js, GraphQL, Qt

Libraries: NumPy, PyTorch, Matplotlib, Apollo, OpenMP, GLM, WebGPU, OpenGL